Eigenvalues and inverse problems

# Finitely many unitary classes with prescribed shifted singular values

**Difficulty:** challenging **Importance:** interesting to the community

**Status: Solved**

**Topic:** Inverse determination from complete pseudospectral data

**Rating rationale:** Generic finiteness follows from invariant theory, but the exceptional fibers resist that argument; the question measures how much complete pseudospectral data can identify a nonnormal matrix.

### Negative resolution - George Stepaniants

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**Solved negatively, 12 September 2026 (UTC).** The Theorem and Sections 1-4 of the complete proof establish a continuous family of complex $9 \times 9$ matrices with the same singular values of every complex scalar shift, with no two distinct family members unitarily similar. Each matrix has nilpotent Jordan type $(3, 3, 3)$. Thus no finite $M_9$ exists, refuting the universal finiteness statement below. This does not assert a classification of the remaining dimensions.

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The full proof passed a separate independent Codex-agent audit. Substantial AI assistance is disclosed; this is informal automated review, not external human peer review or Lean verification. The proof uses an exact block determinant identity and a polynomial dimension argument, with the standard constant-rank theorem. Its finite diagnostic checks are not the proof. Fortier Bourque and Ransford retain credit for the question and generic finiteness theorem, which is compatible with this exceptional fiber.

**Retained historical material.** The original ratings, context, exact target, references and dated pre-resolution status-search text are preserved below. The earlier generic result is unchanged; the older references to an unresolved question describe the pre-resolution record.

### Context and notation

For a complex $N \times N$ matrix $M$, write $s_1(M) \geq \cdots \geq s_N(M)$ for its singular values. Say $A \sim_{\text{sip}} B$ if

$$s_j(A - zI) = s_j(B - zI) \qquad (z \in \mathbb{C},\ 1 \leq j \leq N).$$

This is equality of *super-identical pseudospectral* data: it includes every singular value of every scalar shift.

### Problem statement

Is it true that for every $N \geq 1$ there is an integer $M_N \geq 1$ such that every family $A_1, \dots, A_{M_N+1} \in \mathbb{C}^{N\times N}$ with $A_i \sim_{\text{sip}} A_j$ for all $i, j$ contains a pair satisfying

$$A_j = U^* A_i U \quad \text{for some } i < j \text{ and } U^* U = I?$$

Equivalently, is every such data fiber a union of at most $M_N$ unitary similarity classes, with a bound depending only on $N$?

### References


- Maxime Fortier Bourque and Thomas Ransford, *Super-identical pseudospectra*, *Journal of the London Mathematical Society* **79**(2), 511–528 (2009), **Theorem 1.4 and §6.2**. The theorem removes a closed measure-zero exceptional set; the question asks to remove this exception. Author manuscript.
- Thomas Ransford, *Pseudospectra and matrix behaviour*, **Theorem 5.4 and the following discussion, pp. 10–11**. This restates the finiteness theorem and explicitly leaves the empty-exceptional-set case open. It explains why replacing algebraicity with integrality fails to prove it.
- G. Armentia, J. M. Gracia and F. E. Velasco, *Identical pseudospectra of any geometric multiplicity*, *Linear Algebra and its Applications* **436**(6), 1683–1688 (2012). Their similarity theorem for super-identical pseudospectra gives ordinary similarity, which allows nonunitary changes of basis and does not answer the displayed question.
- T. Ransford and N. Walsh, *A four-mean theorem and its application to pseudospectra*, arXiv:2109.14472v2, 9 July 2022, Theorems 1.3–1.4. Polynomial-norm comparisons and similarity-conditioning bounds do not establish finiteness of the unitary classes; the related norm question is retained in MF-24.


The original low-dimensional results give $M_2 = 1$ and $M_3 = 2$; $M_1 = 1$ is immediate. The unresolved question concerns exceptional data fibers in higher dimensions, rather than generic matrices.

### Scope and status check

The quantifiers above are an explicit restatement of the source question about the exceptional set, with $M_N$ counting classes instead of the source's pigeonhole family size. The source is older: current openness rests on bounded later-literature searches, not a recent explicit reaffirmation. Searches through 2026-09-11 for super-identical pseudospectra, finite unitary classes, the exceptional set and related similarity results found no resolution. The 2012 ordinary-similarity theorem and the cited 2022 norm estimates were checked for scope. They do not remove the exceptional set.